



Quiz 10.03 (Note: Paper Quiz. Nothing to upload. Click for Review Guide)

Completion requirements

Due: Friday, October 3, 2025, 10:20 AM

DSC 360 — Study Guide for Quiz (≈15 minutes)

What to expect

Short, concept-focused check on big ideas since the last quiz. No heavy math; explain in your own words.

1. Key Terms (define + give a 1-line example)

- **Corpus** — the body of documents you are retrieving from.
- **Semantic embeddings** — dense vectors capturing meaning, not just keywords.
- **Chunking** — splitting long docs into smaller pieces so embeddings capture focused meaning.
Why overlap? Sometimes a sentence or idea straddles a boundary; a small overlap ensures context isn't lost.
- **Cosine similarity / cosine distance** — measures angle (direction of meaning) not length.
- **K-nearest neighbors (KNN)** — search method: find closest embeddings to a query.
- **Hit@K** — fraction of queries where the correct answer is in the top K retrieved.
- **Hallucinations** — model invents content when retrieval fails; mitigate with guardrails + context checks.

2. RAG: the Pipeline

1. **Ingest** — collect docs → chunk → embed → index
2. **Retrieve** — embed query → KNN search → filter/threshold
3. **Construct context** — select, format, fit within token budget
4. **Generate** — prompt design, temperature control
5. **Guardrails** — “use only context,” similarity cutoff, option to decline
6. **Evaluate** — Hit@K + qualitative rubric

Retrieval sources may be local files, SQL, APIs/web search, or a vector DB. Retrieval is swappable; the rest of the loop stays the same.

3. How text becomes vectors (encodings)

Older methods (covered in Socratica video):

1. **Bag of Words (BoW)** — counts word frequencies; ignores order; sparse, huge vectors.
2. **TF-IDF** — reweights BoW by word rarity; still sparse, orderless.

Modern method (what we use):

3. **Neural (transformer-based) embeddings** — pretrained models (e.g. **qwen3-embedding:8b**, Lab 04; smaller embedding model, Lab 03) map text into dense semantic spaces where geometry reflects meaning.
 - Work for words, sentences, or whole documents.
 - Power cosine similarity + KNN retrieval in modern RAG.

4. KNN Search

Once we have embeddings, we need a way to find the most relevant chunks.

- **K-nearest neighbors (KNN) search** finds the K closest vectors to the query embedding.
- “Closeness” is usually measured with **cosine similarity** (angle between vectors).
- Example: if K=3, the system retrieves the 3 most semantically similar chunks.
- KNN is simple but effective — the backbone of retrieval in RAG.
- Note: This is the same underlying idea as KNN classification/regression (which use neighbors to assign a label or value). In our course, we apply KNN specifically to **search** for similar text embeddings.

5. Evaluation and Pitfalls

Evaluation

- *Quantitative*: Hit@1/3/5, latency (embed vs. generate), simple accuracy on a test set.
- *Qualitative rubric*: correctness, idiomaticity, specificity, clarity, citations.
- *Safety*: if top similarity is low → respond “insufficient context” (don’t guess).

Common pitfalls

- Oversized chunks (dilutes signal)
- No overlap → context split across chunks
- Too-low cutoff (noisy context)
- Prompts that don't restrict to retrieved context
- Evaluating only Hit@K (misses answer quality)

6. Prep Checklist + Videos

Sample prompts (practice explaining):

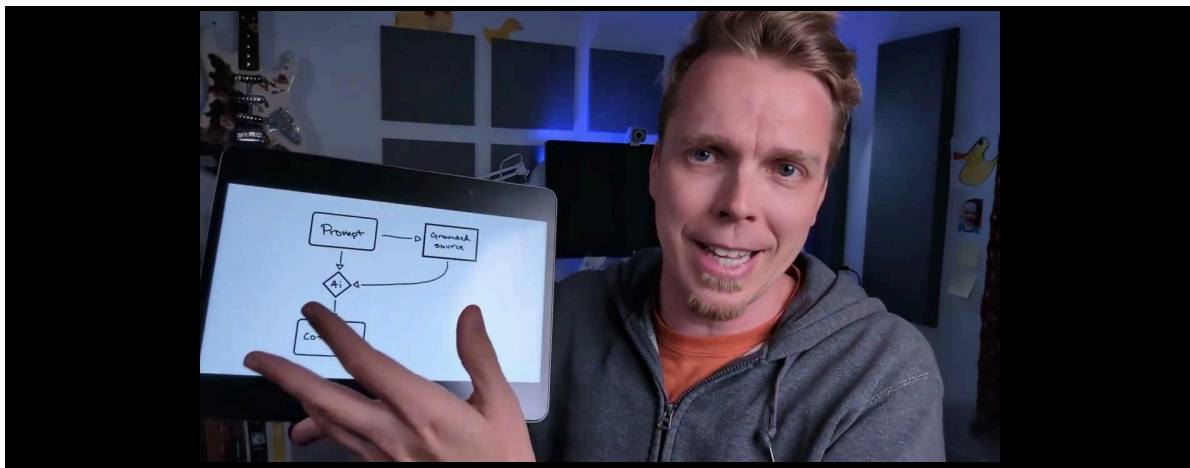
- "Why do we chunk before embedding?"
- "How does Hit@K differ from answer accuracy?"
- "Given a low top similarity, what should a safe RAG system do?"

Checklist:

- Review Lab 03/04 notes on retrieval and prompting.
- Be ready to sketch the RAG loop and describe cosine similarity/KNN in words.
- Watch the three short videos below.

AI Explained: What is RAG — Retrieval Augmented Generation? — Morten Rand-Hendriksen

Short (3 min) explainer. Focus: grounding LLMs with external sources, how RAG improves accuracy and traceability, and the role of **semantic cache**.

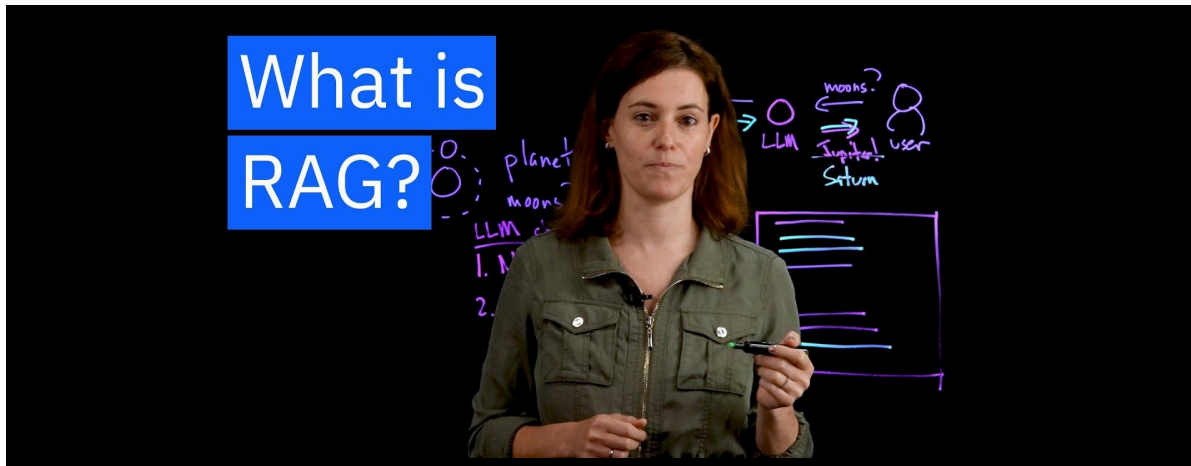


Play Video

▶ AI Explained: What is RAG - Retrieval Augmented Generation?

What is Retrieval-Augmented Generation (RAG)? – IBM Technology

Focus: the retrieve → augment → generate loop; how RAG reduces hallucinations and keeps answers current.



Play Video

What is Retrieval-Augmented Generation (RAG)?

Cosine Similarity (NLP) – Socratica

Focus: cosine similarity vs. distance; “angle not length”; why this underpins KNN retrieval.

Note: This video demonstrates Bag of Words. In our labs we use **neural transformer embeddings**, which are more powerful and semantically meaningful.

Ignore the Wolfram Alpha part. What matters is the idea of cosine similarity and the **similarity matrix**. That’s essentially what we did in Labs 03 and 04, except that we multiplied the embeddings matrix of the corpus by a single vector (the embedded search query) to get a vector of similarity scores.



Play Video

 [Cosine Similarity](#) ← [Natural Language Processing](#) ← [Socratica](#)